

CISC3024 Pattern Recognition

Homework 1

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3.1

$\omega_1 = \{[0,0]^T, [0,1]^T\}$, $\omega_2 = \{[1,0]^T, [1,1]^T\}$, $\rho = 1$, $w(0) = [0,0]^T$.

Augment each feature vector by 1:

$$x = [x_1, x_2, 1]^T, w = [w_1, w_2, w_0]^T$$

Then $g(x) = w^T x$. Set $w(0) = [0,0,0]^T$.

Training vectors:

$$x^{(1)} = [0,0,1]^T, x^{(2)} = [0,1,1]^T \in \omega_1$$

$$x^{(3)} = [1,0,1]^T, x^{(4)} = [1,1,1]^T \in \omega_2$$

Reward-and-punishment form:

If $x \in \omega_1$ and $w^T x \leq 0$, then $w \leftarrow w + \rho x$.

If $x \in \omega_2$ and $w^T x \geq 0$, then $w \leftarrow w - \rho x$.

Otherwise w is unchanged.

Present samples in the order $x^{(1)}, x^{(2)}, x^{(3)}, x^{(4)}$. Repeat until one full cycle has no update.

t=1: $x^{(1)} = [0,0,1]^T \in \omega_1$, $w = [0,0,0]^T$, $w^T x = 0 \leq 0$. $w = [0,0,0]^T + [0,0,1]^T = [0,0,1]^T$.

t=2: $x^{(2)} = [0,1,1]^T \in \omega_1$, $w^T x = 1 > 0$. No update.

t=3: $x^{(3)} = [1,0,1]^T \in \omega_2$, $w^T x = 1 \geq 0$. $w = [0,0,1]^T - [1,0,1]^T = [-1,0,0]^T$.

t=4: $x^{(4)} = [1,1,1]^T \in \omega_2$, $w^T x = -1 < 0$. No update.

t=5: $x^{(1)} = [0,0,1]^T \in \omega_1$, $w^T x = 0 \leq 0$. $w = [-1,0,0]^T + [0,0,1]^T = [-1,0,1]^T$.

t=6: $x^{(2)} = [0,1,1]^T \in \omega_1$, $w^T x = 1 > 0$. No update.

t=7: $x^{(3)} = [1,0,1]^T \in \omega_2$, $w^T x = 0 \geq 0$. $w = [-1,0,1]^T - [1,0,1]^T = [-2,0,0]^T$.

t=8: $x^{(4)} = [1,1,1]^T \in \omega_2$, $w^T x = -2 < 0$. No update.

t=9: $x^{(1)} = [0,0,1]^T \in \omega_1$, $w^T x = 0 \leq 0$. $w = [-2,0,0]^T + [0,0,1]^T = [-2,0,1]^T$.

t=10: $x^{(2)} = [0,1,1]^T \in \omega_1$, $w^T x = 1 > 0$. No update.

t=11: $x^{(3)} = [1,0,1]^T \in \omega_2$, $w^T x = -1 < 0$. No update.

t=12: $x^{(4)} = [1,1,1]^T \in \omega_2$, $w^T x = -1 < 0$. No update.

t=13: $x^{(1)} = [0,0,1]^T \in \omega_1$, $w^T x = 1 > 0$. No update.

t=14: $x^{(2)} = [0,1,1]^T \in \omega_1$, $w^T x = 1 > 0$. No update.

t=15: $x^{(3)} = [1,0,1]^T \in \omega_2$, $w^T x = -1 < 0$. No update.

t=16: $x^{(4)} = [1,1,1]^T \in \omega_2$, $w^T x = -1 < 0$. No update.

One full cycle has no update. The algorithm stops.

The algorithm converges to $w = [-2,0,1]^T$. The separating line is $-2x_1 + w_0 = 0$, i.e. $-2x_1 + 1 = 0$, $x_1 = \frac{1}{2}$.

3.2

The three lines are realized by three perceptrons in the first layer. Activation function:

$$f(v) = \begin{cases} 1, & v \geq 0 \\ 0, & v < 0 \end{cases}$$

$$g_1(x) = x_1 + x_2, y_1 = f(g_1(x))$$

$$g_2(x) = x_2 - \frac{1}{4}, y_2 = f(g_2(x))$$

$$g_3(x) = x_1 - x_2, y_3 = f(g_3(x))$$

Synaptic weights of the first layer, written as $[w_1, w_2, w_0]^T$:

$$w_1 = [1,1,0]^T, w_2 = [0,1,-1/4]^T, w_3 = [1,-1,0]^T$$

Intersection points: $(-1/4, 1/4)$, $(0, 0)$, $(1/4, 1/4)$.

The first layer performs the mapping $x \mapsto y = [y_1, y_2, y_3]^T$, $y_i \in \{0,1\}$. Each region corresponds to one vertex of the unit cube.

region	test point	(g1, g2, g3)	cube vertex y
R_1	$(0, 1/8)$	$(+, -, -)$	$(1, 0, 0)$
R_2	$(0, 1)$	$(+, +, -)$	$(1, 1, 0)$
R_3	$(-2, 1)$	$(-, +, -)$	$(0, 1, 0)$
R_4	$(2, 1)$	$(+, +, +)$	$(1, 1, 1)$
R_5	$(1, -0.2)$	$(+, -, +)$	$(1, 0, 1)$
R_6	$(-1, -0.2)$	$(-, -, -)$	$(0, 0, 0)$
R_7	$(0, -1)$	$(-, -, +)$	$(0, 0, 1)$

Vertex $(0, 1, 1)$ is not used.

(a) Two-layer network

Let $\omega_1 = \{(0,0,0), (1,0,0), (0,1,0), (1,1,0)\}$ and $\omega_2 = \{(0,0,1), (1,0,1), (1,1,1)\}$.

These two sets of vertices are linearly separable by $y_3 = 1/2$. The output perceptron is

$$g(y) = -y_3 + \frac{1}{2} = 0$$

If $y_3 = 0$, then $g = 1/2 > 0$ (ω_1). If $y_3 = 1$, then $g = -1/2 < 0$ (ω_2).

Output synaptic weights $[w_{y_1}, w_{y_2}, w_{y_3}, w_0]^T = [0, 0, -1, 1/2]^T$. First-layer weights are given above.

(b) Three-layer network

Let $\omega_1 = \{(0,0,0), (1,1,1)\}$ and let ω_2 be the remaining five realized vertices. This is the same situation as XOR: one hyperplane in y -space cannot put $(0, 0, 0)$ and $(1, 1, 1)$ on one side and the other vertices on the other side. Therefore a two-layer network is not sufficient.

Second layer, vertex $(0, 0, 0)$:

$$z_1 = f(-y_1 - y_2 - y_3 + \frac{1}{2}), w_{z1} = [-1, -1, -1, 1/2]^T$$

At $(0, 0, 0)$, $g = 1/2 > 0$. At any other vertex, at least one y_i equals 1, so $g \leq -1/2 < 0$.

Second layer, vertex $(1, 1, 1)$:

$$z_2 = f(y_1 + y_2 + y_3 - \frac{5}{2}), w_{z2} = [1, 1, 1, -5/2]^T$$

At $(1, 1, 1)$, $g = 1/2 > 0$. At any other vertex, $y_1 + y_2 + y_3 \leq 2$, so $g \leq -1/2 < 0$.

Mapping: $(0,0,0) \mapsto (z_1, z_2) = (1,0)$, $(1,1,1) \mapsto (0,1)$, other vertices $\mapsto (0,0)$. Output OR:

$$g(z) = z_1 + z_2 - \frac{1}{2}, w = [1, 1, -1/2]^T$$

If z_1 or z_2 equals 1, then $g \geq 1/2 > 0$ (ω_1).

Synaptic weights:

layer	neuron	$w = [\text{weights}, w_0]^T$
1	y_1	$[1, 1, 0]^T$
1	y_2	$[0, 1, -1/4]^T$
1	y_3	$[1, -1, 0]^T$
2	z_1	$[-1, -1, -1, 1/2]^T$
2	z_2	$[1, 1, 1, -5/2]^T$
3	output (OR)	$[1, 1, -1/2]^T$